Cipher Forensics and Attack Recommendation Engine

CMPS455 – Computer Security

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Introduction

Cryptography protects sensitive information

Cryptanalysis studies how encrypted messages can be analysed

Different ciphers leave different patterns

The project analyses ciphertext and estimates the encryption family

Project Objective



Analyse ciphertext patterns



Detect the most likely cipher type



Explain the reasoning behind detection



Recommend attack strategies



Demonstrate basic cryptanalysis concepts

Supported Cipher Types

Caesar Cipher

- Letters shifted by fixed value

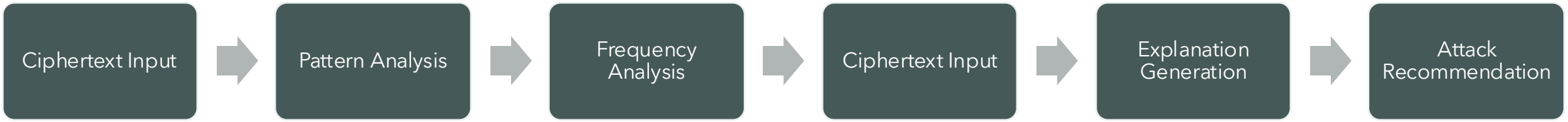
Substitution Cipher

- Letters replaced with other letters

Transposition Cipher

- Letters rearranged without replacement

System Workflow



Frequency Analysis



System counts letter occurrences



English letters have known frequencies



Patterns help identify cipher types

Detection Logic

Condition	Predicted Cipher
Few Unique Letters	Caesar Cipher
Moderate variation	Substitution Cipher
High Variation	Transposition Cipher

Recommended Attack Strategies

Cipher	Recommended Attack
Caesar Cipher	Brute Force
Substitution Cipher	Frequency Analysis
Transposition Cipher	Structural Analysis

Python Implementation

Implemented using Python

Simple modular structure

Functions used:

analyze_frequency()

detect_cipher()

recommend_attack()

Example System Output

```
===Cipher Forensics and Attack Recommendation Engine===
Enter ciphertext: KHOOR ZRUOG

===Pattern Analysis===
Letter Frequency: {'K': 1, 'H': 1, 'O': 3, 'R': 2, 'Z': 1, 'U': 1, 'G': 1}
Number of Unique Letters: 7

===Estimated Encryption Family===
Detected Cipher: Caesar Cipher

===Explanation===
The ciphertext contains a small number of unique letters. This means that it is a simple shift pattern, which is Caesar cipher.

===Recommended Attack Strateg===
Use brute-force attack by trying all 25 possible shifts.
```

Discussion

01

Strengths:

- Simple and understandable
- Demonstrates cryptanalysis concepts
- Combines reasoning and attack planning

02

Limitations:

- Supports only classical ciphers
- Uses simplified detection rules
- No automatic decryption

Future Work

- Add automatic decryption
- Support modern encryption algorithms
- Improve accuracy using AI/ML
- Add graphical user interface
- Expand attack techniques

Conclusion

- System successfully analyses ciphertext
- Detects likely cipher families
- Explains reasoning behind detection
- Recommends suitable attack methods
- Demonstrates core cryptanalysis principles